

errors of system:

- if Meas. devices are not perfect Small error $0,1 \rightarrow 1\%$
But importantly you assume clocks are synchronized
(because we look at same timestep to fix calc power consumption)

- if Clocks is offset then two sensors may claim
to meas. same time interval but physically avg.
over diff. time intervals

ref interval: $[0, T]$ offset interval: $[\delta, T+\delta]$

Meas. deviation:

$$E(T, \delta) := \hat{m}(T, 0) - \hat{m}(T, \delta) \quad ; \quad \hat{m}(T, \delta) := \frac{1}{T} \int_{\delta}^{T+\delta} m(t) dt$$

Markov Modulated Process extends CTMC by attaching
value at each state

$$\underline{P}(t) = \underline{P}(0) \exp(\underline{Q}t)$$

$$\underline{E} = \text{diag}(e_1, e_2, \dots, e_n)$$

Power consumed in
State i

E_{ii}

Discretize Meas values to States